

Olympiad Test Paper — Social Science (Class 7) — Paper 5

Chapter 19: Infrastructure: Engine of India's Development

Paper 5 — (progressively harder)

Subject	Social Science (NCERT Class 7)
Chapter	19 — Infrastructure: Engine of India's Development
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct.

Apex traps: the physical/social split is by FUNCTION (move-and-supply vs build-human-capability), never by cost, visibility, size, age or material; a facility's TYPE never changes just because something else depends on it (a power grid stays physical even when a hospital needs it; a hospital stays social even though it draws power); buried utilities and people-serving institutions are fully infrastructure; the British built the 1853 railways for extraction, trade and troop control and only LATER were they repurposed as a 20-million-a-day national lifeline; care is a shared duty of citizens AND government; J.C. Bose (c.1895) was a communication scientist who improved the coherer and declined patents — he built no railway or highway; tele+communicare and infra+structura roots; aviation=avis; TEU is a container unit, not speed/height/distance; renewable energy parks, living root bridges (Khasi/Jaintia, Meghalaya) and Kautilya's Arthashastra all sit INSIDE the chapter. Do not write on this paper — mark answers on your answer sheet.

1. Four works are funded in one budget: a buried fibre-optic backbone, a children's public park, a sewage-treatment plant with its pipes, and a magistrate's court. A reporter wants the share of the budget items that are PHYSICAL infrastructure. Counting correctly, that share is:

- (A) three of the four (everything except the park)
- (B) one of the four (only the sewage plant, since networks are too invisible to count)
- (C) two of the four (the fibre-optic backbone and the sewage-treatment plant)
- (D) all four, because each was built by the same public budget

2. A student reasons: 'Infra means below and structura means a structure; therefore infrastructure must be whatever is physically buried underground, like pipes and cables.'

The most precise correction is that the roots describe:

- (A) exactly and only what is physically below ground level, so above-ground roads are not truly infrastructure
- (B) structures shared between distant places, since 'infra' means 'between'
- (C) the underlying structure that everything else is built upon, a meaning that figuratively covers visible roads and people-serving schools too, not only literally buried things
- (D) the tallest structures that rise above a city, since 'structura' refers to height

3. A hospital cannot run without electricity, so a debater concludes: 'A hospital is really energy infrastructure once you trace its dependence.' The chapter's classification logic rejects this because:

- (A) a hospital becomes physical infrastructure the moment it consumes power, so the debater is correct
- (B) the power grid becomes social infrastructure because a hospital depends on it
- (C) a facility's type is fixed by the function it itself performs, so the hospital stays social infrastructure even though it depends on the separate, physical energy infrastructure that supplies its power
- (D) neither the hospital nor the grid is infrastructure, since only roads qualify

4. Three openings happen in one week: a new expressway, a new public hospital, and a new piped-water main to homes. Classified by the chapter's two arms with the correct physical sub-type named, they are respectively:

- (A) physical transport, social, physical utility
- (B) physical transport, physical utility, social
- (C) social, physical transport, physical energy
- (D) physical energy, social, physical communication

5. A district has gleaming highways and a busy port, yet daily power cuts and burst water mains, plus too few schools and clinics. A planner must name the SINGLE most complete reason its development still lags. The best answer is that:

- (A) transport, factories and social services all rely on steady power and water, and development also needs healthy, educated people, so failing utilities and weak social infrastructure together drag the district down despite good transport
- (B) its good highways and port are themselves the harm and should be removed to let development begin
- (C) only roads matter, so the failing utilities and missing schools are irrelevant to development
- (D) the district will develop fastest of all, because strong transport infrastructure alone guarantees progress

6. Region P has reliable power but no schools or clinics; Region Q has many schools and clinics but daily blackouts. Following the chapter, the prescription that fits BOTH regions'

actual gaps is:

- (A) P should build more power stations and Q should build more schools, doubling each region's existing strength
- (B) both regions should pour everything into new highways and ignore their real shortfalls
- (C) P should build social infrastructure (schools, clinics) while Q should strengthen its physical energy infrastructure
- (D) neither region needs to act, since one strong arm of infrastructure is enough on its own

7. A factory owner reports that after new roads, a reliable grid and a nearby port opened, his costs fell. The tightest cause-and-effect chain consistent with the chapter is:

- (A) the new infrastructure cut his costs only by forcing him to dismiss most of his workers
- (B) the roads, grid and port raised his costs, because maintaining nearby public works is billed to the factory
- (C) the infrastructure made no difference, because a factory's cost is set purely by the world price of raw materials
- (D) faster roads and the port speed raw materials in and finished goods out, while steady power keeps machines running, so each unit is made and shipped more cheaply and quickly

8. A teacher asks which single sentence distinguishes how each arm of infrastructure aids development WITHOUT any embedded error. The correct sentence is:

- (A) physical infrastructure produces educated people, while social infrastructure moves goods and power
- (B) both arms only move goods, and neither affects the people of a country
- (C) physical infrastructure helps cities alone, while social infrastructure helps villages alone
- (D) physical infrastructure moves goods, power and information, while social infrastructure produces healthy, educated, capable people

9. A debater says: 'Whether a thing is infrastructure depends on how much it cost — cheap things are too minor to count.' The example that most cleanly destroys this is:

- (A) only ports and dams, the most expensive works, are ever called infrastructure
- (B) every item the chapter calls infrastructure happens to cost exactly the same to build
- (C) a low-cost footpath and a very expensive expressway are both transport infrastructure, proving cost does not decide the category
- (D) cheap works are filed as social infrastructure and costly works as physical

10. Match each to its correct category: (i) a coastal wind farm, (ii) a buried water-supply pipeline, (iii) a public legal-aid court, (iv) a 5G mobile-tower network. The fully correct set is:

- (A) (i) physical energy, (ii) social, (iii) social, (iv) physical communication
- (B) (i) social, (ii) physical utility, (iii) physical communication, (iv) social
- (C) (i) physical energy, (ii) physical utility, (iii) social, (iv) physical communication
- (D) (i) physical communication, (ii) physical energy, (iii) social, (iv) physical utility

11. A student claims India's railways and its road network must hold the same world ranking. Using only the chapter's facts, the correct response is:

- (A) yes, a country's road-size rank and railway-passenger rank are always identical
- (B) the chapter states the road network is the world's second-largest and that the railways carry about 20 million people a day, but it never equates a network's size rank with a passenger-load rank, so the two are separate facts
- (C) no, because the chapter says India has no railways at all
- (D) no, because the chapter says India's roads are actually the world's smallest

12. The Golden Quadrilateral is named a 'quadrilateral'. A student guesses it has four lanes throughout. The accurate reason for the name is that it:

- (A) has exactly four traffic lanes along its entire length
- (B) connects four major cities — Delhi, Mumbai, Chennai and Kolkata — forming a four-sided road circuit across the country
- (C) runs through four small villages located on the plains
- (D) joins four foreign capital cities to one another

13. A logistics firm must rush perishable cargo from Delhi to a buyer in Chennai by road. Of the chapter's listed roads, the single best fit and the reason are:

- (A) a city metro line, because it is the fastest way to move freight between distant states
- (B) NH44 alone, because it is described as a short ring road circling only Delhi
- (C) the Golden Quadrilateral, because it gives a fast, direct highway link between those major cities so goods reach the distant market quickly
- (D) a coastal shipping lane, because the Golden Quadrilateral is a sea route between ports

14. NH44 and an expressway are both highlighted as fast long roads, yet a student treats them as identical. The chapter's distinction is best captured by saying:

- (A) NH44 is a single city's underground metro line while an expressway is a national road
- (B) NH44 is a sea port and an expressway is a power line, so they share no category
- (C) NH44 is social infrastructure while an expressway is energy infrastructure
- (D) both are physical transport roads built by the central government to link distant parts of the country; NH44 is named as one of India's major and longest national highways in particular

15. A student writes: 'The British built the 1853 railways as India's lifeline, so they have served ordinary Indians from day one.' The most accurate correction is that the railways:

- (A) did carry ordinary Indian travellers as a lifeline from their very first day in 1853
- (B) were never a lifeline and still serve only British interests today
- (C) were built in 1853 by Indians offering free public travel to all
- (D) were first built for British colonial ends — exporting raw materials, selling British goods and moving troops — and only LATER were repurposed into a national lifeline carrying millions of Indians

16. The chapter uses the railways' history to teach a general principle. Which fresh scenario best illustrates the SAME principle?

- (A) a road that is torn up the moment its original builder leaves, because old works must always be destroyed
- (B) a fort wall built centuries ago to defend a kingdom is now preserved and used as a popular public heritage park — infrastructure built for one purpose later serving a very different, wider one
- (C) a bridge that becomes useless within a few years of being built
- (D) a port that can only ever be used for the single cargo it first handled and nothing else

17. A pupil misreads '20 million passengers a day' as '20 thousand a day'. By roughly what factor has he understated the true figure?

- (A) about one hundred times too small (B) about ten times too small
- (C) about one million times too small (D) about one thousand times too small

18. A commuter complains city roads jam every morning. The chapter's reasoning recommends a metro because it:

- (A) moves large numbers of commuters on rails within the city, taking pressure off the congested roads
- (B) exports the city's raw materials to foreign sea ports
- (C) generates renewable electricity to power the whole city
- (D) carries cargo containers between the country's two coasts

19. A student says a metro and an expressway 'do the same job the same way'. The clearest distinction the chapter supports is that:

- (A) a metro carries cargo ships while an expressway carries aircraft
- (B) a metro is social infrastructure while an expressway is energy infrastructure
- (C) a metro carries commuters on rails within a single city, while an expressway is a high-speed road linking cities across states
- (D) both are inter-state roads, differing only in the colour of their road signs

20. The word 'aviation' comes from the Latin avis ('bird'). Extending this root consistently, aviation infrastructure centres on:

- (A) harbours and ships that carry cargo across the sea
- (B) airports and aircraft that carry people and goods through the air

- (C) power stations that generate electricity for the grid
- (D) pipelines that carry clean water to homes

21. India is said to have about 159 airports. A student concludes India must therefore have exactly 159 sea ports as well. The correct response is:

- (A) yes, a country's airport count and sea-port count are always equal
- (B) no, because the chapter states India actually has no airports at all
- (C) no — 159 is the chapter's airport figure, and sea ports are a separate kind of coastal facility that the chapter does not give the same number
- (D) no, because all 159 of those facilities are in fact sea ports, not airports

22. Port A handles 5,000 TEUs and Port B handles 8,500 TEUs on the same day. A second day, Port A rises to 9,000 TEUs while Port B holds steady. Comparing each port's two-day totals, the correct reading is:

- (A) Port A handled more, since its single best day of 9,000 beats Port B's best day of 8,500
- (B) the comparison cannot be made, because TEU measures how fast each port's cranes move, not how many containers
- (C) Port B is taller in structure by 3,000 metres, which is what the higher TEU total really means
- (D) Port A's two-day total of 14,000 standard containers exceeds Port B's 17,000 only if you misadd — in fact Port B's 17,000 is greater than Port A's 14,000, so Port B handled more standard twenty-foot-equivalent containers over the two days

23. Why does the world use a standard unit like the TEU when comparing ports of very different sizes and cargo mixes?

- (A) it provides a common container measure, so ports handling different goods and volumes can be compared fairly on the same scale
- (B) it records the exact height of each port's tallest loading crane
- (C) it counts how many airports each port happens to own nearby
- (D) it measures how fast the metro trains run inside each port city

24. India's major sea ports sit on both the eastern and western coasts. The chapter's geographic reasoning for why this matters to trade is that:

- (A) coastal ports are useless, since ships can never reach the country's interior
- (B) ports must instead be built in the Himalayas to handle the heaviest cargo
- (C) ports on both coasts automatically double the country's number of airports
- (D) ports on both coasts give India sea access in two directions, easing both imports and exports to and from different parts of the world

25. A history quiz states: 'J.C. Bose designed and built India's first railway in 1853.' The single fact that makes this wrong is that:

- (A) Bose built the railway, but in 1947 rather than in 1853
- (B) Bose was a communication scientist active around 1895, whereas the 1853 railways were a British colonial project, not his work
- (C) the first railway was actually built by Indians for free travel, with Bose only assisting
- (D) Bose designed the Golden Quadrilateral road network rather than the railway

26. Order these three chapter dates earliest to latest: J.C. Bose's wireless work (about 1895), the first Indian Railways (1853), Indian independence (1947). The correct sequence is:

- (A) 1853 railways, then 1895 wireless work, then 1947 independence
- (B) 1895 wireless work, then 1853 railways, then 1947 independence
- (C) 1947 independence, then 1853 railways, then 1895 wireless work
- (D) 1853 railways, then 1947 independence, then 1895 wireless work

27. A 'coherer', improved by J.C. Bose, performed which role in early wireless communication?

- (A) it detected and decoded the transmitted radio signals at the receiving end
- (B) it generated the electricity that powered the early railways
- (C) it pumped clean water into the homes of a growing city
- (D) it carried passengers along an elevated urban rail track

28. The chapter notes Bose 'chose not to patent much of his work', and defines a patent as a temporary exclusive right. The most reasonable inference about Bose, plus the consequence had he patented everything, is:

- (A) he believed his inventions were worthless, and patenting them would have made them work
- (B) he was willing to let his ideas be used freely rather than seek exclusive personal profit; had he patented all of it, then for a number of years only he or those he permitted could legally make, sell or use those inventions
- (C) he was legally barred from patents, and patenting would have let everyone copy his work freely
- (D) he wanted his findings kept secret forever, and patenting would have automatically published them to all

29. Telecommunication comes from tele ('far') and communicate ('to share'). By this exact root meaning, which activity qualifies as telecommunication?

- (A) driving a truck of goods along a national highway between two cities
- (B) loading shipping containers onto a vessel at a busy sea port
- (C) generating electricity from the wind at a coastal wind farm
- (D) phoning a relative who lives in a distant city to share news with them

30. Communication infrastructure and transport infrastructure both 'connect' distant places. The clearest distinction the chapter draws is that:

- (A) communication infrastructure moves information across distance, while transport infrastructure moves people and goods
- (B) communication moves goods, while transport moves only information
- (C) both move only electricity and nothing else between places
- (D) communication works only abroad, while transport works only inside India

31. Energy infrastructure includes dams, solar parks and wind farms. Sorting these by the source each taps, the correct statement is:

- (A) all three can supply renewable energy — flowing water for the dam, sunlight for the solar park, and wind for the wind farm
- (B) all three burn coal to produce their energy
- (C) only the dam is energy infrastructure, while the others are social
- (D) none of them is infrastructure, because nature supplies the energy for free

32. A city retires a coal power plant and replaces it with a large solar park feeding the same grid. In the chapter's terms, the city's energy infrastructure becomes more:

- (A) sustainable, because it now draws on a cleaner, renewable source that harms the environment less
- (B) polluting, because a solar park burns more fuel than a coal plant
- (C) social rather than physical, because the public will use the power
- (D) useless, because renewable energy cannot power a whole city

33. The chapter argues energy and utilities matter to ALL other infrastructure. The reasoning that supports this is:

- (A) energy and utilities are used only inside private homes and nowhere else
- (B) energy and utilities entirely replace the need for roads and schools
- (C) transport, communication and social services all need power and water to function, so energy and utilities underpin every other arm
- (D) energy and utilities are completely unrelated to any other infrastructure

34. A 'utility' takes its name from Latin utilis ('useful'). The reason the name suits water, electricity and gas supply is that they:

- (A) are basic useful services supplied to homes and workplaces for everyday needs
- (B) are the most beautiful structures built anywhere in a city
- (C) are the fastest roads available for long-distance travel
- (D) are devices that detect and decode wireless signals

35. Sanitation is listed under social infrastructure. The strongest reason it belongs there, rather than being dismissed as minor, is that:

- (A) sanitation makes aircraft take off and fly noticeably faster
- (B) sanitation raises the container count handled at a sea port

- (C) sanitation adds extra kilometres to the national highway network
- (D) clean sanitation protects people's health and wellbeing, which is a core aim of social infrastructure

36. The chapter links social infrastructure to development through people. The most complete statement of this link is:

- (A) schools and hospitals produce educated and healthy citizens, who can then work, invent and contribute to the country's growth
- (B) social infrastructure speeds up cargo trains, raising the country's exports
- (C) social infrastructure entirely replaces the need for any physical infrastructure
- (D) social infrastructure benefits only the people who actually build it

37. A vandalised public bus shelter sparks a debate. Student 1: 'Only the government must fix and protect it.' Student 2: 'Citizens share that duty too.' The chapter supports:

- (A) Student 1, because citizens have no role at all in public property
- (B) Student 1, because only foreigners are meant to maintain Indian infrastructure
- (C) neither, because public infrastructure needs no care of any kind
- (D) Student 2, because caring for public infrastructure is a collective responsibility shared by citizens and government together

38. Which set of citizen actions best demonstrates collective responsibility for shared infrastructure?

- (A) breaking the taps in a public washroom when no one is watching
- (B) ignoring a leaking public pipeline because it is 'not my job'
- (C) waiting for a foreign agency to come and repair the local road
- (D) using a public park properly, not damaging the benches, and reporting a broken streetlight to be fixed

39. A village treats its hand-pump, school building and approach road as 'everyone's to look after'. The civic principle this reflects is:

- (A) private ownership, where each facility belongs to a single family
- (B) collective responsibility — shared public infrastructure cared for jointly by the whole community
- (C) government monopoly, where citizens are barred from touching any of it
- (D) foreign maintenance, where outsiders are expected to care for all of it

40. A living root bridge in Meghalaya and a concrete flyover are both infrastructure. The living root bridge most clearly illustrates that infrastructure can be:

- (A) only ever made of concrete and steel to be durable
- (B) sustainable and in harmony with nature, grown by guiding tree roots across a stream as the Khasi and Jaintia tribes do
- (C) always harmful to the surrounding environment
- (D) impossible to build without modern machines

- 41.** Sustainable infrastructure, as the chapter describes it, is built so as to:
- (A) use clean energy and eco-friendly materials, minimising pollution and harm to biodiversity
 - (B) use the cheapest, most polluting materials available to save money
 - (C) ignore the environment completely so the work can be finished fastest
 - (D) rely only on ancient hand tools no matter what the project needs
- 42.** Kautilya's Arthashastra is cited in the chapter. Its mention is meant to show that:
- (A) concern for roads and public works is an ancient Indian idea, not only a modern one
 - (B) infrastructure was completely unknown in India until British rule
 - (C) only Western thinkers ever wrote about public works
 - (D) the Arthashastra was composed after the Golden Quadrilateral was built
- 43.** A summary card reads: 'Infrastructure is only the visible roads and buildings, the British built the 1853 railways to help Indians, and caring for it is the government's job alone.' How many of these three claims are wrong?
- (A) only one is wrong
 - (B) only two are wrong
 - (C) none is wrong
 - (D) all three are wrong
- 44.** Reviewing the whole chapter, the single FULLY correct statement, with no embedded error, is:
- (A) infrastructure is only physical, and the 1853 British railways were built to benefit Indian travellers
 - (B) infrastructure, the engine of development, has physical and social arms — including invisible utilities and natural works like Meghalaya's living root bridges — and is best kept working through care shared between government and citizens
 - (C) infrastructure means only invisible systems, and its upkeep is solely the government's responsibility
 - (D) infrastructure includes social services, but J.C. Bose built India's first railway in 1853
- 45.** A school principal labels six items: (i) the school library, (ii) the road outside the gate, (iii) the electricity line powering the classrooms, (iv) the local court, (v) the staff drinking-water tank fed by a municipal main, (vi) the school's broadband router fed by a fibre network. How many are SOCIAL infrastructure?
- (A) three — the library, the court and the drinking-water tank
 - (B) two — the library and the court
 - (C) four — the library, the court, the water tank and the broadband router
 - (D) one — only the library
- 46.** A district masterplan lists six projects: a hydroelectric dam, a public hospital, a national expressway, a sewage-treatment utility, a community sports park, and a mobile-

tower network. The number that are PHYSICAL infrastructure is:

- (A) three — only the dam, the expressway and the sewage utility
- (B) five — everything except the hospital
- (C) two — only the dam and the expressway
- (D) four — the dam, the expressway, the sewage utility and the mobile-tower network

47. A reporter says, 'India's railways must rank second-largest in passengers because its road network is second-largest in size.' The flaw in this reasoning is best named as:

- (A) treating a fact about one kind of infrastructure (road-network size) as if it must transfer to a different measure of another (railway passenger numbers), when the chapter links the two by no such rule
- (B) denying that India has any railways, when the chapter clearly mentions them
- (C) confusing kilometres of road with the number of airports India has
- (D) assuming the railways are still British-owned and so cannot be ranked

48. A town wants ONE project that will most directly relieve morning road jams, and a SECOND that will most directly raise the share of children finishing school. The best matched pair is:

- (A) a solar park for the road jams and a sea port for the children
- (B) more schools for the road jams and a metro for the children
- (C) a metro for the road jams (physical transport) and more schools for the children (social infrastructure)
- (D) a water pipeline for the road jams and a wind farm for the children

49. A planner notes: 'Our coal plant, our buried water main and our buried fibre cable are all underground or out of sight, so all three are the same type of infrastructure.' The single error in this reasoning is that:

- (A) being out of sight does not set the type — by function the coal plant is energy, the water main is a utility, and the fibre cable is communication, three different physical sub-types
- (B) all three are actually social infrastructure, because the public ultimately benefits
- (C) none of the three is infrastructure, because infrastructure must always be visible
- (D) all three are the same type, since visibility is exactly how the chapter sorts infrastructure

50. Two facilities are linked: a metro depot is powered entirely by a nearby solar park. A student says this 'merges' them into one type. The correct view is:

- (A) the solar park becomes transport infrastructure because it serves a metro
- (B) they remain two distinct types — the metro depot is physical transport and the solar park is physical energy — even though the transport facility depends on the energy one
- (C) the metro depot becomes energy infrastructure because it runs on solar power
- (D) both become social infrastructure because commuters benefit from the link

51. A teacher draws a chain: good roads and ports cut a factory's costs, the cheaper goods sell better, the factory hires more workers, and those workers' children attend the new town school. The chapter's framework reads this chain as:

- (A) proof that only physical infrastructure matters, since the school appears only at the very end
- (B) proof that social infrastructure alone caused the factory's lower costs
- (C) physical infrastructure boosting economic activity, whose gains then meet social infrastructure when the workers' children are educated — both arms reinforcing development
- (D) proof that infrastructure has no effect, since the chain has too many steps to be real

52. A claim states: 'Because the 1853 railways were British and exploitative, India should have destroyed them at independence.' Using the chapter's railway principle, the soundest reply is:

- (A) yes, colonial infrastructure must always be torn down once the colonisers leave
- (B) the railways should have been kept but used only for British-style goods export
- (C) infrastructure built for one purpose can be repurposed for a wider one, which is exactly what happened — the colonial railways became a national lifeline carrying millions of Indians daily
- (D) the railways were useless after 1947 and faded away within a few years

53. A quiz lists three Indian contributions to communication science and asks which is correctly described. The accurate one is:

- (A) J.C. Bose built India's first national highway in 1853 and patented the design
- (B) J.C. Bose pioneered radio-wave (wireless) communication research and improved the coherer around 1895, and chose not to patent much of his work
- (C) J.C. Bose designed the Golden Quadrilateral road network and the TEU container
- (D) J.C. Bose invented the railway in 1853 and later founded India's first sea port

54. A planner argues a new airport and a new sea port are 'totally different infrastructure with nothing in common'. The chapter's most accurate correction is that, while their modes differ:

- (A) both are actually social infrastructure, because travellers are people
- (B) the airport is communication infrastructure and the port is energy infrastructure
- (C) both belong to the same broad category of physical transport infrastructure, since each moves people and goods, one by air and one by sea
- (D) they share nothing, because only roads count as transport infrastructure

55. A student insists 'invisible things and people-services can't both be infrastructure — pick one rule.' The chapter's actual rule that admits BOTH a buried gas pipeline and a public hospital is:

- (A) only things you can see and touch are infrastructure, so the pipeline is excluded
- (B) only things that serve people directly are infrastructure, so the pipeline is excluded

- (C) only the most expensive of the two can be infrastructure, so just one qualifies
- (D) anything that is a basic facility a country needs to function is infrastructure, split by function into physical (like the pipeline) and social (like the hospital), regardless of whether it is seen or who uses it

56. A debate motion reads: 'Energy infrastructure is the least important arm because power plants don't carry people or teach children.' The strongest rebuttal from the chapter is:

- (A) energy is most important because it is the only true infrastructure of all
- (B) transport, communication and social services all depend on power and water to run, so energy and utilities underpin every other arm rather than being least important
- (C) energy does not matter, because trains and hospitals can run with no power
- (D) energy is purely social infrastructure, so the motion misclassifies it

57. A municipality replaces vandal-prone open wiring with a sustainably built, eco-friendly streetlight system and asks citizens to report faults. This single project advances which TWO chapter ideas at once?

- (A) the colonial origin of the railways, and the meaning of the word aviation
- (B) the second-largest road network, and the chronology of J.C. Bose's wireless work
- (C) sustainable infrastructure built with eco-friendly methods, and the collective civic duty of citizens to help care for shared infrastructure
- (D) the definition of a TEU, and the four cities of the Golden Quadrilateral

58. A pupil ranks four facilities by 'how directly each builds people's capabilities': a public university, a power transmission line, a children's vaccination clinic, and a container port. From most to least direct on that specific measure, the order is:

- (A) the power line and the port rank highest, because they cost the most to build
- (B) the university and the clinic rank highest as social infrastructure building people's learning and health, while the transmission line and the container port are physical infrastructure that build capability only indirectly
- (C) all four rank equally, because every facility builds people's capabilities the same way
- (D) the port ranks highest, because moving containers educates the workforce

59. A summary states: 'The British built the 1853 railways to extract resources, J.C. Bose patented all his wireless inventions for profit around 1895, and infrastructure care is the government's job alone.' Exactly how many of these three claims are correct?

- (A) all three are correct
- (B) only one — that the British built the 1853 railways for extraction
- (C) exactly two are correct
- (D) none is correct

60. A class is told to find the ONE statement that is entirely accurate about how India's transport infrastructure both serves the nation today and reflects its history. The fully correct statement is:

(A) the railways were a national lifeline from 1853, while highways and airports are social infrastructure serving travellers

(B) metros link distant states while the long-distance railways serve only single cities, and all of these are energy infrastructure

(C) the Golden Quadrilateral joins four foreign capitals and the railways carry about 20 people a day across India

(D) highways like the Golden Quadrilateral and NH44, metros, airports and the railways are all physical transport infrastructure, and the railways in particular began as a colonial tool in 1853 before becoming a national lifeline carrying about 20 million people a day

Solutions — Social Science (Class 7), Chapter 19: Infrastructure: Engine of India's Development — Paper 5

Paper 5 — (progressively harder)

Marking: +4 correct, -1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	C	11	B	21	C	31	A	41	A	51	C
2	C	12	B	22	D	32	A	42	A	52	C
3	C	13	C	23	A	33	C	43	D	53	B
4	A	14	D	24	D	34	A	44	B	54	C
5	A	15	D	25	B	35	D	45	B	55	D
6	C	16	B	26	A	36	A	46	D	56	B
7	D	17	D	27	A	37	D	47	A	57	C
8	D	18	A	28	B	38	D	48	C	58	B
9	C	19	C	29	D	39	B	49	A	59	B
10	C	20	B	30	A	40	B	50	B	60	D

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (C) Physical infrastructure covers transport, utilities, communication and energy. The fibre-optic backbone is physical (communication) and the sewage-treatment plant is physical (utility) — two items. The children's park and the magistrate's court are people-serving social infrastructure (recreation and justice), so the count is two, not three. The 'only the sewage plant' answer wrongly demotes the buried network for being invisible, but the split is by function, not visibility, so a communication network still counts. 'All four' confuses funding source with category — sharing a budget does not make a court or a park physical.

2. (C) Infra ('below') + structura ('a structure') give the figurative sense of the foundational base supporting everything built on top — it is not a literal demand that the thing sit under the soil. Highways (above ground) and schools (people-serving) are still infrastructure. The 'only physically below ground' reading takes the metaphor literally and

wrongly excludes roads. 'Shared between distant places' misdefines infra as 'between' (that is the tele/inter idea, not infra). 'Tallest structures' inverts the meaning entirely, since infra is 'below', not 'above'.

3. (C) Classification follows what a facility does: a hospital serves people's health (social), and a power grid supplies energy (physical). Dependence does not transfer category — the hospital draws on energy infrastructure yet remains social itself. Saying the hospital 'becomes' energy infrastructure confuses using power with generating it. Saying the grid 'becomes social' inverts the same error in the other direction. Treating only roads as infrastructure ignores both arms — social services and energy systems are both basic facilities a country needs.

4. (A) An expressway moves people and goods, so it is physical transport; a hospital serves health, so it is social; a piped-water main supplies a basic service to homes, so it is a physical utility. The order is therefore physical transport, social, physical utility. The option that ends 'utility, social' swaps the hospital and the water main. The 'physical energy' label misnames water supply as energy. 'Physical communication' misnames water as a communication network — water is a utility, not a signal carrier.

5. (A) Development needs all arms working together: utilities (power, water) underpin transport, industry and social services, and social infrastructure (schools, clinics) builds the capable people development depends on. With both failing, good transport cannot compensate — that is the complete reason. Removing the highways and port is absurd; they are assets, not harms. 'Only roads matter' ignores the very gaps causing the lag. 'Develop fastest of all' contradicts the chapter's point that physical infrastructure alone does not guarantee progress.

6. (C) Each region must fix the arm it lacks: P already has power but lacks people-serving services, so it needs social infrastructure; Q has services but unreliable power, so it needs energy infrastructure. The reversed answer tells each region to add more of what it already has, which leaves the real gap open. Pouring everything into highways addresses neither gap. 'Neither needs to act' contradicts the chapter's claim that both physical and social arms are needed for full development.

7. (D) Better transport shortens supply and delivery times, and reliable power prevents costly stoppages, so production and distribution both get cheaper — that is the chapter's reasoning. The 'dismiss workers' answer invents a layoff mechanism the chapter never claims. 'Raised his costs' contradicts the reported fall and wrongly bills public works to the factory. 'No difference' denies the very effect the owner observed and ignores how infrastructure shapes production costs.

8. (D) Physical systems carry goods, power and information; social services build people's health and skills — that pairing is correct. The first wrong option swaps the two roles. 'Both only move goods' erases the human-development role of social infrastructure

entirely. The 'cities versus villages' split is an invented geographic divide the chapter never draws; both arms matter everywhere.

9. (C) The split is by function, not price: a cheap footpath and a costly expressway both move people, so both are transport infrastructure — cost is irrelevant. 'Only the priciest works count' restates the very error being refuted. 'Everything costs the same' is plainly false and is not how the chapter divides infrastructure. Filing by cheap-versus-costly invents a cost-based split that the chapter explicitly does not use.

10. (C) A wind farm generates power (energy); a water pipeline supplies a basic service (utility); a court delivers justice to people (social); a mobile-tower network carries signals (communication). Only the first set matches all four. The second wrongly calls the water pipeline social — supplying water is a utility, not a people-serving service. The third miscategorises the wind farm as social and the court as communication. The fourth swaps the energy and communication labels onto the wrong items.

11. (B) The chapter gives two distinct facts — second-largest road network and roughly 20 million rail passengers daily — and never links the two as the same ranking, so the claim is unsupported. 'Always identical' invents a rule the chapter does not state. 'No railways at all' contradicts the 20-million figure. 'Smallest roads' flatly reverses the stated second-largest network.

12. (B) The name comes from its four-cornered shape linking four big cities into a four-sided loop, not from a lane count. The 'four lanes' guess confuses geometry of the route with the number of lanes. 'Four small villages' is wrong — it links major metros, not villages. 'Four foreign capitals' moves the network outside India entirely, which is false.

13. (C) The Golden Quadrilateral directly connects Delhi and Chennai (two of its four corner cities) by fast highway, ideal for rushing road cargo between them. A metro is an intra-city commuter rail, not an inter-state freight road. NH44 is a major long national highway, not a short Delhi ring road. The Golden Quadrilateral is a road network, not a sea route, so the coastal-lane reasoning is false.

14. (D) Both NH44 and expressways are high-speed national roads under the central government, with NH44 noted as among India's longest highways — so they share a category and differ only in specifics. Calling NH44 a metro line misnames a national highway as urban rail. Calling it a sea port and the expressway a power line places them in wrong categories entirely. Labelling NH44 social and the expressway energy ignores that both are physical transport.

15. (D) The 1853 railways served colonial extraction, trade and control first; their role as a mass Indian lifeline came later — that is the repurposing the chapter teaches. 'Lifeline from day one' is exactly the misconception being corrected. 'Still serve only the British'

ignores the 20-million-a-day Indian ridership today. 'Built by Indians for free travel' is historically false; the British built them for their own purposes.

16. (B) The railway principle is that infrastructure built for one purpose can be repurposed for a different, often wider, need — exactly mirrored by an old defensive wall reused as a public park. 'Torn up when the builder leaves' contradicts the principle, since the railways were kept, not destroyed. 'Useless within a few years' denies durability and repurposing. 'Only ever one cargo' is the opposite of the flexibility the principle describes.

17. (D) Twenty million is twenty thousand multiplied by one thousand, so reading it as twenty thousand understates the figure roughly one-thousand-fold. 'One hundred times' would turn twenty million into two hundred thousand, not twenty thousand. 'Ten times' gives two million, far too large. 'One million times' would shrink it to twenty, which is the different misread the earlier papers used; here the misread is to twenty thousand, a factor of about a thousand.

18. (A) A metro is a high-capacity urban rail system; by carrying crowds on rails it relieves road congestion — the chapter's exact point. Exporting raw materials is a port-and-rail freight role, not a metro's. Generating electricity is the work of energy infrastructure, not a metro. Carrying containers between coasts describes ships and ports, not an urban commuter rail.

19. (C) A metro is intra-city rail; an expressway is inter-state high-speed road — different modes and different reach, though both are physical transport. 'Cargo ships' and 'aircraft' belong to ports and aviation, not these two. Labelling a metro social and an expressway energy mistypes both; both are physical transport. 'Both are inter-state roads' is wrong because a metro is rail confined to one city.

20. (B) Avis ('bird') points to flight, so aviation infrastructure is airports and aircraft moving people and goods by air. Harbours and ships are sea transport, a different mode. Power stations are energy infrastructure. Pipelines are utilities. Only the air-transport answer follows the bird root.

21. (C) The 159 figure is for airports; sea ports are distinct coastal facilities with no matching number stated, so the equality is invented. 'Always equal' is a fabricated rule. 'No airports at all' contradicts the 159 figure itself. 'All 159 are sea ports' relabels airports as ports, which is false.

22. (D) TEU counts standard twenty-foot-equivalent containers. Over two days Port A handled $5,000 + 9,000 = 14,000$ and Port B handled $8,500 + 8,500 = 17,000$, so Port B handled more in total — the larger total, not the single best day, settles 'more containers'. Comparing only best days (9,000 vs 8,500) ignores the full two-day totals and gives the wrong winner. TEU is a container count, not crane speed, so the 'cannot be made' answer

misdefines the unit. TEU has nothing to do with a port's height, so the '3,000 metres taller' reading is nonsense.

23. (A) A shared unit like the TEU lets unlike ports be ranked on one fair scale of container throughput. Crane height, airport ownership and metro speed are unrelated quantities that a container-handling unit does not and could not measure.

24. (D) Two coastlines mean sea routes opening in two directions, widening trade access — the chapter's point. 'Useless coastal ports' is self-contradictory, since ports exist precisely to connect sea and land. The Himalayas are landlocked mountains, the worst place for a sea port. Coastal ports have no bearing on airport counts, so the 'double the airports' answer is unrelated.

25. (B) Bose worked on wireless communication around 1895; the 1853 railways were built by the colonial British, so the claim is doubly off — wrong man, wrong field. Changing the date to 1947 keeps the false claim that he built the railway. 'Built by Indians for free travel' is historically false. 'Designed the Golden Quadrilateral' assigns him a modern road project he had nothing to do with.

26. (A) Numerically 1853 precedes 1895, which precedes 1947, so railways come first, Bose's wireless work next, and independence last. Putting 1895 before 1853 reverses the first two. Putting 1947 first ignores that independence is the latest of the three. Placing 1895 after 1947 wrongly makes the wireless work the most recent event.

27. (A) A coherer was a receiver-side device that picked up and decoded incoming radio-wave signals, which is why Bose's improvement mattered to communication. Generating railway electricity is an energy-infrastructure function. Pumping water is a utility's job. Carrying passengers on rails is a metro's role. None of these is what a coherer did.

28. (B) Declining patents suggests a willingness to let science be freely used; and since a patent grants exclusivity, patenting everything would have legally reserved making and selling to him alone for a set period. 'Worthless inventions' contradicts a respected scientist's pioneering work, and patents do not make things work. 'Legally barred' is unsupported, and patents restrict rather than open up copying. Declining patents is not the same as secrecy, and a patent is not a publishing mechanism.

29. (D) Tele + communicare means sharing information across a distance, which a long-distance phone call does exactly. Driving goods is transport, not information-sharing. Loading containers is sea freight. Generating wind power is energy production. Only the phone call sends information far, fitting the roots.

30. (A) The chapter splits them by what each carries: communication carries information, transport carries people and goods. The first wrong option swaps the two payloads. 'Both only move electricity' is false for both. The 'abroad versus inside India' divide is invented; both operate everywhere.

31. (A) Hydro, solar and wind all draw on renewable natural sources — water, sun and wind respectively. 'All burn coal' is false; none of these three is a coal plant. Calling solar parks and wind farms 'social' mistypes energy works as people-serving services. 'Free natural energy means not infrastructure' ignores that the dams, panels and turbines are built facilities that qualify as infrastructure.

32. (A) Swapping coal for solar shifts to a clean, renewable source, making the energy infrastructure more sustainable. 'More polluting' is backwards; solar burns no fuel. The switch does not change category — generating power stays physical energy infrastructure regardless of who uses it. 'Useless' contradicts the chapter's positive treatment of solar parks as real energy infrastructure.

33. (C) Power and water are inputs everything else depends on — trains, signals, hospitals all need them — so energy and utilities underpin the whole system. 'Only inside private homes' ignores their use across all infrastructure. 'Replace roads and schools' overstates their role; they support rather than substitute. 'Completely unrelated' is the opposite of the dependency the chapter describes.

34. (A) Utilis ('useful') fits everyday essential services like water, power and gas delivered to homes and workplaces. Beauty is irrelevant to the name. Fast roads are transport, not utilities. Devices that decode wireless signals are coherers, a communication item, not a utility.

35. (D) Sanitation safeguards health, a central goal of social infrastructure, so it is rightly grouped there. It has nothing to do with aircraft speed, port container counts or highway length — those three answers attach sanitation to unrelated physical facilities.

36. (A) Social infrastructure builds capable, healthy, educated people who then drive growth — the full chain. Speeding cargo trains is a transport effect, not a social-infrastructure one. 'Replaces physical infrastructure' is false; both arms are needed. 'Benefits only its builders' ignores that schools and hospitals serve the whole population.

37. (D) The chapter frames upkeep as a shared duty of citizens and government, so Student 2 is right. 'Citizens have no role' contradicts that shared duty. 'Only foreigners maintain it' is baseless. 'Needs no care' denies the very premise that infrastructure must be looked after to keep serving everyone.

38. (D) Proper use, avoiding damage and reporting faults are exactly the citizen behaviours the chapter calls collective responsibility. Breaking taps is the opposite. Ignoring a leak as 'not my job' rejects the shared duty. Waiting on a foreign agency hands off a duty that belongs to citizens and government together.

39. (B) Treating shared facilities as everyone's to maintain is the chapter's collective responsibility. Private ownership would assign each to one family, the opposite of shared.

'Government monopoly' barring citizens contradicts the shared duty. 'Foreign maintenance' offloads the duty abroad, which the chapter rejects.

40. (B) A living root bridge is grown from tree roots, showing infrastructure can be natural, sustainable and nature-friendly. 'Only concrete and steel' is refuted by the bridge itself. 'Always harmful to the environment' is the opposite of what this eco-friendly example shows. 'Impossible without machines' is disproved by a bridge grown by hand over years.

41. (A) Sustainable infrastructure deliberately uses clean energy and eco-friendly materials to limit environmental and biodiversity harm. Choosing the most polluting materials is the opposite goal. Ignoring the environment for speed defeats sustainability. Insisting only on ancient hand tools confuses 'sustainable' with 'primitive' — the aim is low harm, not avoiding modern methods.

42. (A) The Arthashastra reference shows India thought about public works long ago, making infrastructure an ancient concern. 'Unknown until the British' is contradicted by this very ancient text. 'Only Western thinkers' ignores Kautilya, an Indian thinker. The Arthashastra is ancient, so claiming it came after a modern highway is chronologically impossible.

43. (D) Each claim is false: infrastructure also includes invisible utilities and social services (not only visible roads and buildings); the British built the 1853 railways for their own colonial ends, not to help Indians; and care is a shared duty, not the government's alone. So all three are wrong, not one, two or none.

44. (B) The correct option states both arms, includes invisible and natural works, and names shared care — all accurate. The first wrong option falsely limits infrastructure to physical and misstates the colonial railways. The second falsely narrows it to invisible systems and the government alone. The third correctly mentions social services but then wrongly credits Bose with the 1853 railway, which was a British project, not his.

45. (B) Social infrastructure serves people's learning, justice, health and recreation: the library (learning) and the court (justice) qualify — two items. The road is transport, the electricity line is energy, the water tank is a utility, and the broadband router is communication; all four are physical. Counting the water tank as social mistakes a utility for a service. Adding the broadband router compounds that error. Counting only the library forgets the court.

46. (D) Physical infrastructure spans energy, transport, utilities and communication: the dam (energy), expressway (transport), sewage-treatment plant (utility) and mobile-tower network (communication) make four. The hospital (health) and sports park (recreation) are social. Counting three omits the communication network. Counting five wrongly adds

the sports park to physical. Counting two omits both the utility and the communication network.

47. (A) The error is importing one infrastructure fact (road size) into an unrelated measure (rail passengers) with no stated link — the chapter never equates the two rankings. The reporter does not deny railways exist; he over-extends a road fact. Airports are not part of his claim, so confusing roads with airports is not the flaw. Ownership is irrelevant to the ranking error he actually makes.

48. (C) Road congestion is a transport problem a metro relieves; school completion is a social-infrastructure aim met by building schools — so the pairing is metro then schools. A solar park (energy) and sea port (freight) address neither problem. Swapping schools and the metro misassigns each to the wrong problem. A water pipeline and wind farm are utility and energy works that do not target congestion or schooling.

49. (A) Visibility never decides type; function does, so these split into energy, utility and communication — three distinct physical sub-types, not one. Calling all three social ignores that they move and supply things rather than build human capability. 'Must be visible to count' contradicts the chapter's inclusion of buried utilities. 'Visibility is how the chapter sorts' is exactly the false premise being corrected; the chapter sorts by function.

50. (B) Dependence does not merge categories: the metro depot stays transport and the solar park stays energy, each typed by its own function. The park does not 'become' transport just because it powers one; nor does the depot 'become' energy by consuming power. Neither turns social merely because commuters benefit — they are both physical infrastructure of different sub-types.

51. (C) The chain runs from physical infrastructure (roads, ports) cutting costs and lifting activity, into social infrastructure (the school) educating the next generation — both arms working together, the chapter's core message. 'Only physical matters' ignores the school's role at the chain's end. 'Social alone cut the costs' is backwards; the cost cut came from roads and ports. 'No effect' denies the very development the chain traces.

52. (C) The railway principle is repurposing: a tool built for colonial extraction became India's lifeline, so destroying it would have wasted a usable asset. 'Always tear down colonial works' contradicts that very principle. Keeping it only for British-style export ignores its transformation into mass passenger service. 'Useless and faded away' is refuted by the 20-million-a-day figure today.

53. (B) Bose's real contribution was wireless research and improving the coherer around 1895, plus declining most patents — that is the accurate description. He built no national highway in 1853, and patenting contradicts his actual choice. He did not design the Golden Quadrilateral or invent the TEU container. He did not invent the railway or found a sea port; those mix him up with British colonial projects.

54. (C) Airports and sea ports are both physical transport infrastructure, differing only in mode (air versus sea). Calling them social mistakes who uses them for what they do — both move goods and people, a transport function. Labelling the airport communication and the port energy mistypes both. 'Only roads are transport' wrongly narrows transport to one mode, excluding air and sea.

55. (D) The unifying rule is 'basic facility a country needs', divided by function into physical and social — admitting both the buried pipeline (physical utility) and the hospital (social). 'Only visible things' wrongly drops the buried pipeline. 'Only direct people-services' also wrongly drops the pipeline. 'Only the most expensive' invents a cost test the chapter never uses; both qualify.

56. (B) Energy underpins everything — trains, signals, hospitals all need power and water — so it is foundational, not least important; that is the rebuttal. 'Only true infrastructure' overcorrects and denies the other arms. 'Trains and hospitals run with no power' is plainly false. 'Energy is purely social' misclassifies physical energy infrastructure and does not rebut the motion.

57. (C) An eco-friendly build embodies sustainable infrastructure, and inviting citizens to report faults embodies collective responsibility — two chapter ideas in one project. Railway origins and aviation's meaning are unrelated to a streetlight project. The road-network ranking and Bose's dates do not bear on it either. TEU and the Golden Quadrilateral concern ports and highways, not streetlights or civic care.

58. (B) Social infrastructure (a university and a vaccination clinic) builds people's skills and health most directly, while a power line and a container port are physical infrastructure that help only indirectly — so the two social facilities top the capability measure. Cost does not measure capability-building, so ranking by price is wrong. The four are not equal on this measure; social ones are more direct. A port moves containers, not knowledge, so it does not top a capability ranking.

59. (B) Only the first claim is true: the 1853 railways did serve British extraction and control. The second is false — Bose chose NOT to patent much of his work, the opposite of patenting all for profit. The third is false — care is a shared duty of citizens and government. So exactly one is correct, not all three, not two, and not none.

60. (D) The correct statement classes highways, metros, airports and railways as physical transport and accurately traces the railways from 1853 colonial origins to a 20-million-a-day lifeline. The first wrong option falsely calls highways and airports social and dates the lifeline role to 1853. The second swaps the roles of metros and long-distance railways and mislabels them all energy. The third puts the Golden Quadrilateral abroad and shrinks ridership to 20 people a day.